

Rajasekhar Gullapalli

Data Engineer

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Summary

Results-driven Data Engineer with 3+ years of expertise in designing and implementing scalable data pipelines and ETL workflows. Proficient in leveraging cloud platforms, big data tools, and databases to transform, integrate, and optimize large datasets for actionable insights. Experienced in ensuring data quality, governance, and security while collaborating with cross-functional teams to meet business requirements. Adept at optimizing performance and utilizing modern data processing and storage solutions, enabling efficient handling of high-volume, real-time analytics across various industries.

Technical Skills

- Programming Languages:** Python (Pandas, NumPy), SQL, Bash
- Big Data Technologies:** Apache Spark, Hadoop, Azure HDInsight, Azure Databricks
- Cloud Platforms:** Azure, AWS, GCP
- Databases:** Azure SQL Database, SQL Server
- ETL Tools:** Apache Airflow, Azure Data Factory
- Data Warehousing:** Azure Synapse Analytics
- Data Processing:** Data Lakes, Data Warehouses
- Machine Learning:** Azure Databricks (Feature extraction, data transformation)
- Data Governance:** Azure Data Lake (Access control, Encryption)
- Tools & Technologies:** SQL Queries (Joins, Subqueries, Window Functions), Data Security, Data Integration

Professional Experience

- Data Engineer, Capital One

01/2024 – Present | Remote, USA

  - Worked on requirement gathering and stakeholder engagement to design and develop the Customer 360 Data Integration Platform. Utilized AWS Glue to orchestrate the flow of customer data from multiple sources into a centralized data warehouse.
  - Leveraged Apache Spark to process large volumes of customer data in real-time, ensuring efficient transformation and integration across different data sources. Integrated Spark jobs with Amazon Redshift for scalable processing.
  - Implemented Hadoop-based data storage solutions using Amazon EMR for distributed data processing, allowing efficient management and retrieval of large customer data sets in a fault-tolerant, scalable environment to support advanced analytics.
  - Created complex SQL queries to perform data aggregation, filtering, and joining from various customer data tables in Amazon RDS, optimizing query performance with indexing and partitioning for large-scale data processing.
  - Used AWS SageMaker to run machine learning models on large customer datasets, performing data engineering tasks such as feature extraction, cleaning, and transformation, contributing to predictive analytics for personalized customer insights.
  - Ensured data security and governance by implementing AWS Lake Formation's RBAC and encryption mechanisms, maintaining compliance with regulatory norms while enabling authorized users to access critical customer data securely.
- Associate Data Engineer, Dell

01/2021 – 12/2022 | Remote, India

  - Designed and implemented the ETL Pipeline for Retail Sales using Apache Airflow, orchestrating tasks to extract, transform, and load sales data into Azure Data Warehouse, ensuring accurate and timely data delivery for business intelligence and reporting.
  - Utilized Python libraries (Pandas, NumPy) to develop transformation scripts, cleaning and processing raw sales data by removing duplicates, filling missing values, and optimizing performance, thereby ensuring high data quality and consistency for downstream analysis.
  - Developed SQL queries to transform and aggregate sales data in an Azure SQL Database, employing complex joins, window functions, and subqueries to derive insights, ensuring high-performance querying and efficient data processing at scale.
  - Integrated Azure services (Data Lake Storage, Data Factory, Synapse Analytics) to store, process, and analyze large volumes of sales data, ensuring data ingestion and transformations, while optimizing for cost and performance in the cloud environment.
  - Collaborated with cross-functional teams to gather data requirements and ensure the ETL pipeline aligned with business objectives, enabling actionable insights into retail sales performance and facilitating decision-making processes across marketing, finance, and operations teams.
  - Monitored and optimized pipeline performance, finding bottlenecks and fine-tuning task execution using Apache Airflow's task dependencies and Python-based transformations for minimal downtime and ensuring the pipeline's scalability and resilience.
  - Ensured security and data governance by implementing Azure Data Lake access controls and encryption, keeping data lineage, and holding to organizational policies for compliance for integrity and security of retail data throughout the ETL process.

Education

- Master of Science, Missouri State University  
Computer Science

01/2023 - 12/2024 | Springfield, MO, USA
- Bachelors of Engineering, Anna University  
Electronics and Communications

06/2018 - 05/2022 | Chennai, India

Certifications

- AWS Certified Solutions Architect
- Databricks Certified data engineer
- Data Science Certificate
- Microsoft Azure data fundamentals (DP-900)